

ANKIT MEHTA

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EDUCATION

Georgia Institute of Technology Aug 2026 – May 2028
M.S. Robotics, School of Interactive Computing: Concentration in Embodied AI & Mechanics

Georgia Institute of Technology Aug 2019 – May 2023
B.S. Computer Science: Concentration in Artificial Intelligence & Information Networks | GPA: 3.79, Highest Honors

RESEARCH

Robot Autonomy and Interactive Learning (RAIL) Lab | Dr. Sonia Chernova Jul 2026 – Current

- Built a **VLM task planner** in ROS 2 for long-horizon household tasks on a Hello Robot Stretch 4 and a Fauna Robotics Sprout humanoid. The planner makes VLM queries at run time from the task, camera images, and memory, then composes a fixed library of eight primitives (navigate, pick up, put down, observe, and four analytic calls), and calls a skill only after an object grounder puts the target in view and in reach. [Link](#)
- Added **runtime failure detection and recovery**: monitor nodes at three rates (10 Hz telemetry, 1 Hz grounding, 0.2 Hz vision) check skill stop predicates and cancel a skill the moment it fails. Memory keeps successful plans and named failure modes with their recoveries. [Link](#)
- Integrated all Stretch 4 sensor data across Zenoh and ROS 2 DDS (RTPS), and **cut image-processing latency by 180 ms**.

Borg Lab | Dr. Frank Dellaert Jul 2026 – Current

- Implement a **hierarchical factor graph (STAG) in GTSAM** for autonomous mobile robots in a warehouse. Three levels carry learned latent states: fleet orchestration and task allocation, single-robot missions on the aisle graph, and motion and perception. Each level passes a sub-goal down and an encoded window up. [Link](#)
- Implement a **three-layer hierarchical control architecture** on continuous and discrete vector spaces for motion planning, trajectory planning, and feedback control. [Link](#)
- Built the warehouse simulator in **NVIDIA Isaac Sim**: pallet trucks and counterbalance AMRs drive through palletizing, staging, drop, marshalling, and outbound dock zones. The scene supplies ground truth, wheel commands, and RTX sensor data.

Intelligent Vision & Automation Lab May 2026 – Aug 2026

- Replaced a Kalman filter with a **GRU velocity predictor** for a non-holonomic robot, trained on 922K samples with new sector aspect-ratio and density rate-of-change features. [Link](#)
- In dynamic gap-based navigation, the GRU predicts gap congestion and **cut reactive planning failures by 30% across 350 episode runs**. Started to train a top-level behavior manager with imitation learning.

EXPERIENCE

Full Stack Software Engineer II | NCR Voyix Jul 2023 – Jul 2026

- Built tree-based hierarchical caching systems for role management UIs, and **cut onboarding time by 20%**.
- Led software for a TypeScript payments integration for point-of-sale systems in APAC and EMEA, with reusable distributed uber-JARs for downstream clients.
- Added facial recognition and ID pre-validation to self-checkout, and **cut average queue wait time by 23.6 s**.

Software Engineering Intern | Cypress.io Aug 2022 – Dec 2022

- Built Run All Specs (Vue, fuzzy sort) and In-App Cloud Integration (GraphQL, URQL, Docker). [Link 1](#), [Link 2](#), [Link 3](#)
- Built a nightly build system that tests new Cypress binaries with GitHub Actions. [Link](#)

PROJECTS

Swarm Robotics: Search & Rescue Jun 2026 – Current

- Built a two-quadruped search-and-rescue system in PyBullet with **no central controller or shared memory**. It uses PID/gait locomotion control, IR ray-cast perception, and a UDP protocol with covariance-intersection belief fusion for potential-field navigation and area search. [Link](#)

LLM Trading & Financial Chatbot Sep 2022 – Mar 2026

- Compared LLM trading-agent performance on different time-series data. Built an NLP portfolio-management chatbot for sector classification, stock recommendations, and market-data charts. The portfolio strategy **won first prize in the 2023 BlackRock Challenge**. [Link 1](#), [Link 2](#)

PUBLICATIONS

Mehta, A. et al. (2022). *Hate Speech and Offensive Language Detection in Ethiopian Languages*. International Conference on Learning Representations (ICLR) Workshop. arxiv.org/abs/2208.09941

TID Lab (2021 – 2023): built a Naïve Bayes language classifier in fastText for low-resource Ethiopian languages, applied graphSAGE to classify inflammatory users, and BERT for post similarity. [Link](#)

SKILLS

Languages & Tools: Python, C++, Java, TypeScript, JavaScript, Golang, C, React, Vue, Docker, Kubernetes, GCP, GraphQL, Git

Robotics & ML: ROS 2, GTSAM, Isaac Sim, MuJoCo, PyBullet, PyTorch, TensorFlow, scikit-learn, OpenCV, SAM2, Hugging Face

Concepts: Embodied AI, Task Planning, Factor Graphs, State Estimation, Navigation, Imitation & Reinforcement Learning, NLP